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Asari

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(54) **ELECTRIC POWER DEVICE AND MOVING OBJECT WITH ELECTRIC POWER DEVICE**

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(51) **Int. Cl.**

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H01R 4/34	(2006.01)
H01R 4/58	(2006.01)
H02G 3/16	(2006.01)

(52) **U.S. Cl.**

CPC ***H02G 3/16*** (2013.01); ***B60R 16/0238*** (2013.01); ***H01R 4/34*** (2013.01); ***H01R 4/58*** (2013.01)

(58) **Field of Classification Search**

CPC B60R 16/0238

See application file for complete search history.

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(57) **ABSTRACT**

An electric power device for a moving object includes a direct current circuit and a support board. The support board electrically insulates a first circuit portion arranged on a first surface and a second circuit portion arranged on a second surface of the direct current circuit. A set of terminals consisting of a pair of a positive connection terminal and a negative connection terminal is arranged on the same side surface of the support board.

8 Claims, 10 Drawing Sheets

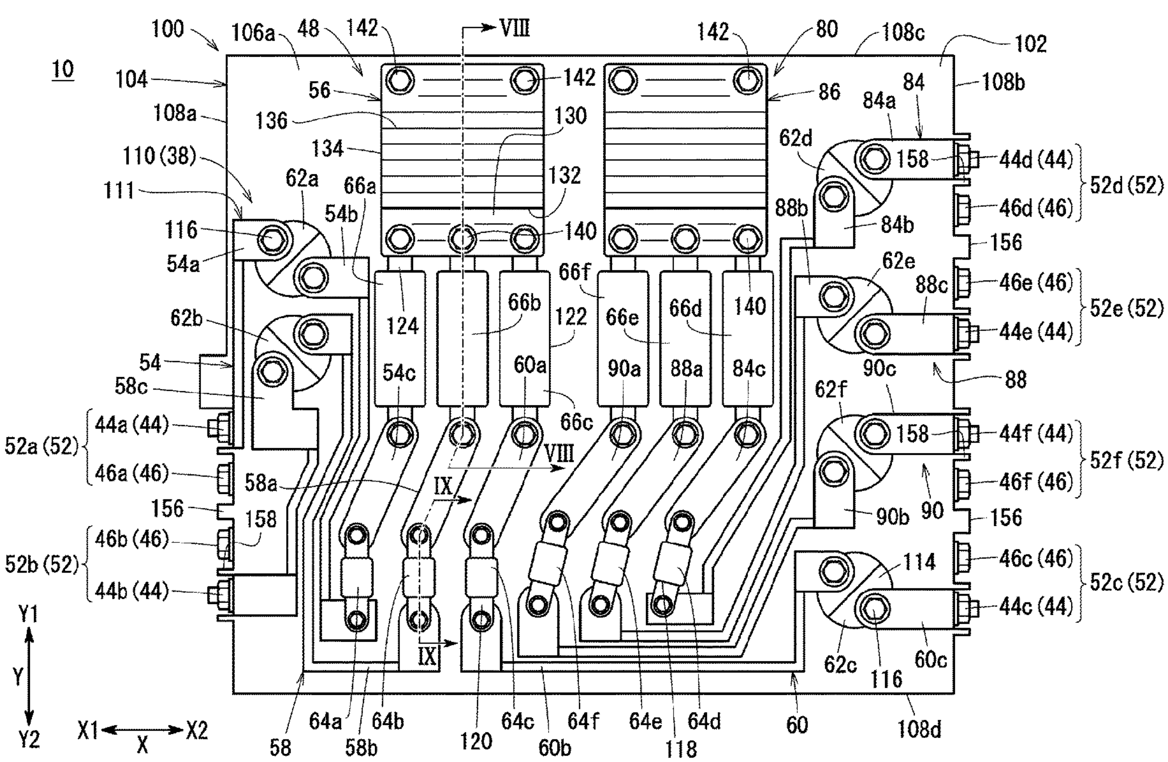

FIG. 1

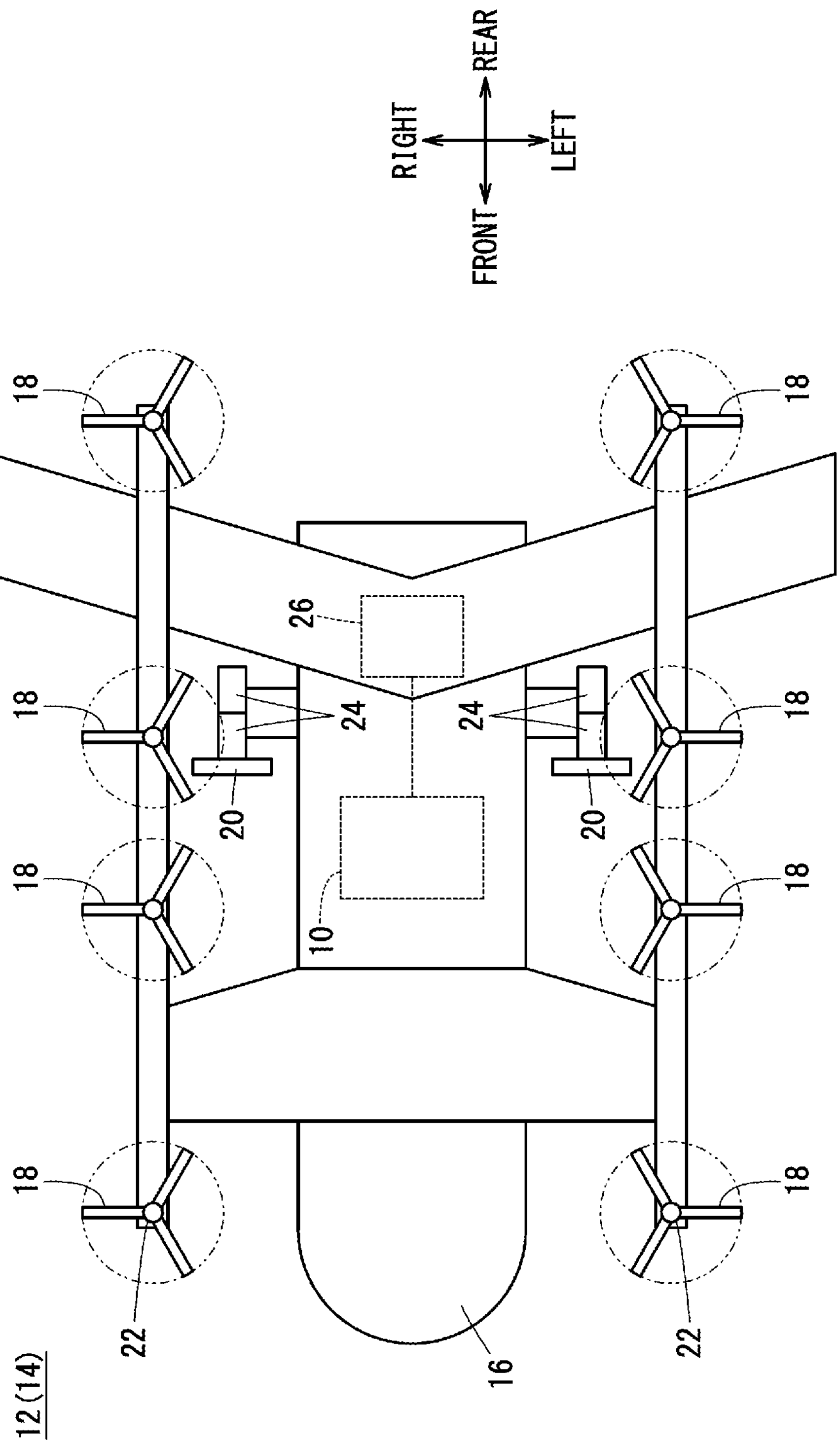
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FIG. 2

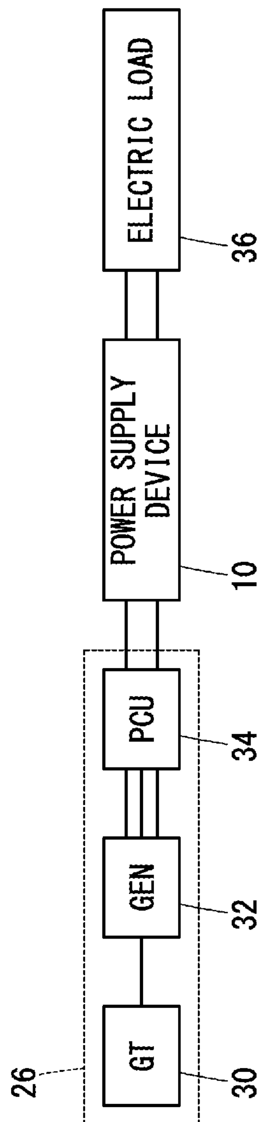
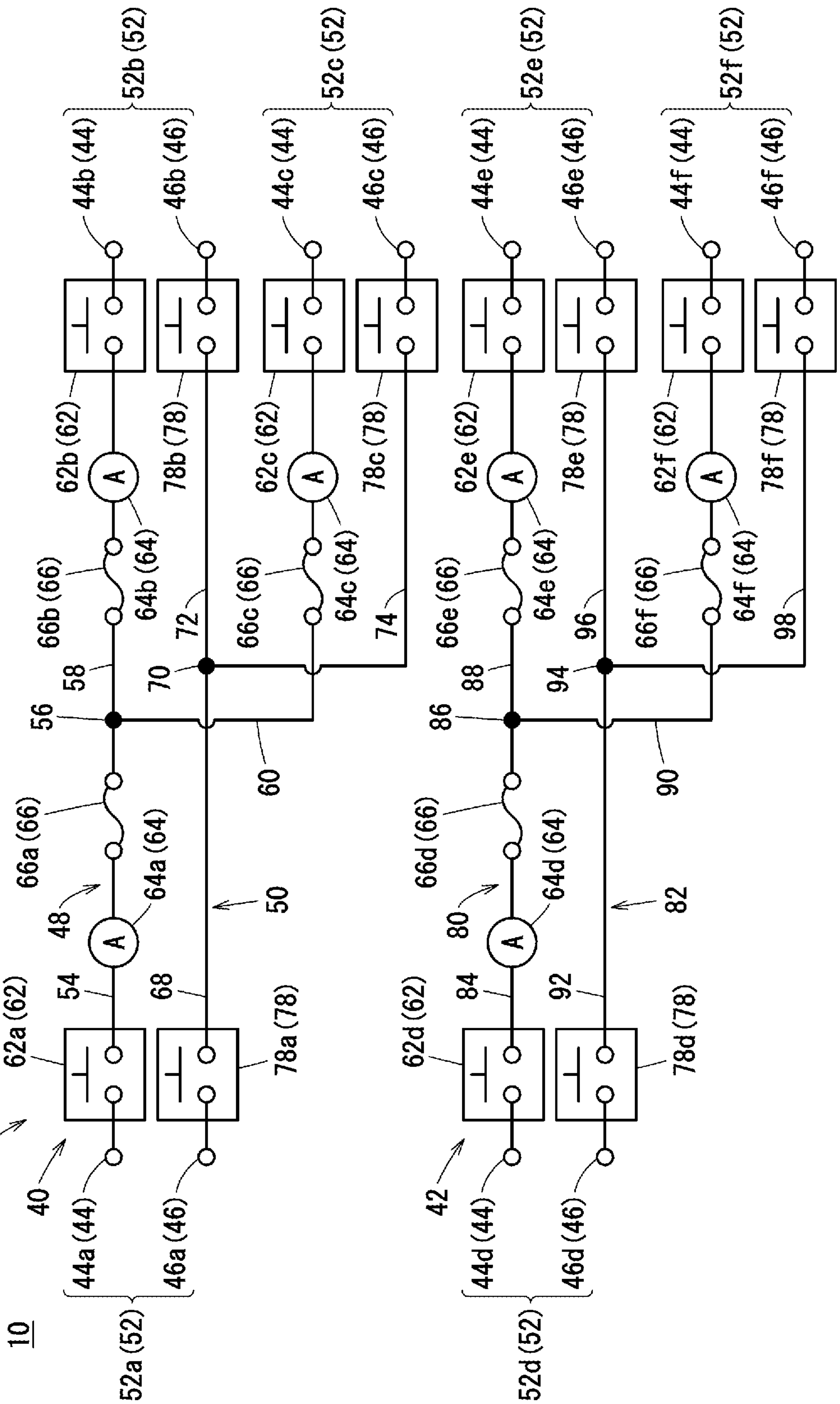


FIG. 3



10
38
40
42
44a (44)
46a (46)
52a (52)
44d (44)
46d (46)
52d (52)
62a (62)
78a (78)
62d (62)
78d (78)
54
68
84
92
48
50
80
82
64a (64)
66a (66)
64d (64)
66d (66)
56
58
60
70
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74
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90
94
96
98
66b (66)
64b (64)
66c (66)
64c (64)
66e (66)
64e (64)
66f (66)
64f (64)
62b (62)
78b (78)
62c (62)
78c (78)
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44e (44)
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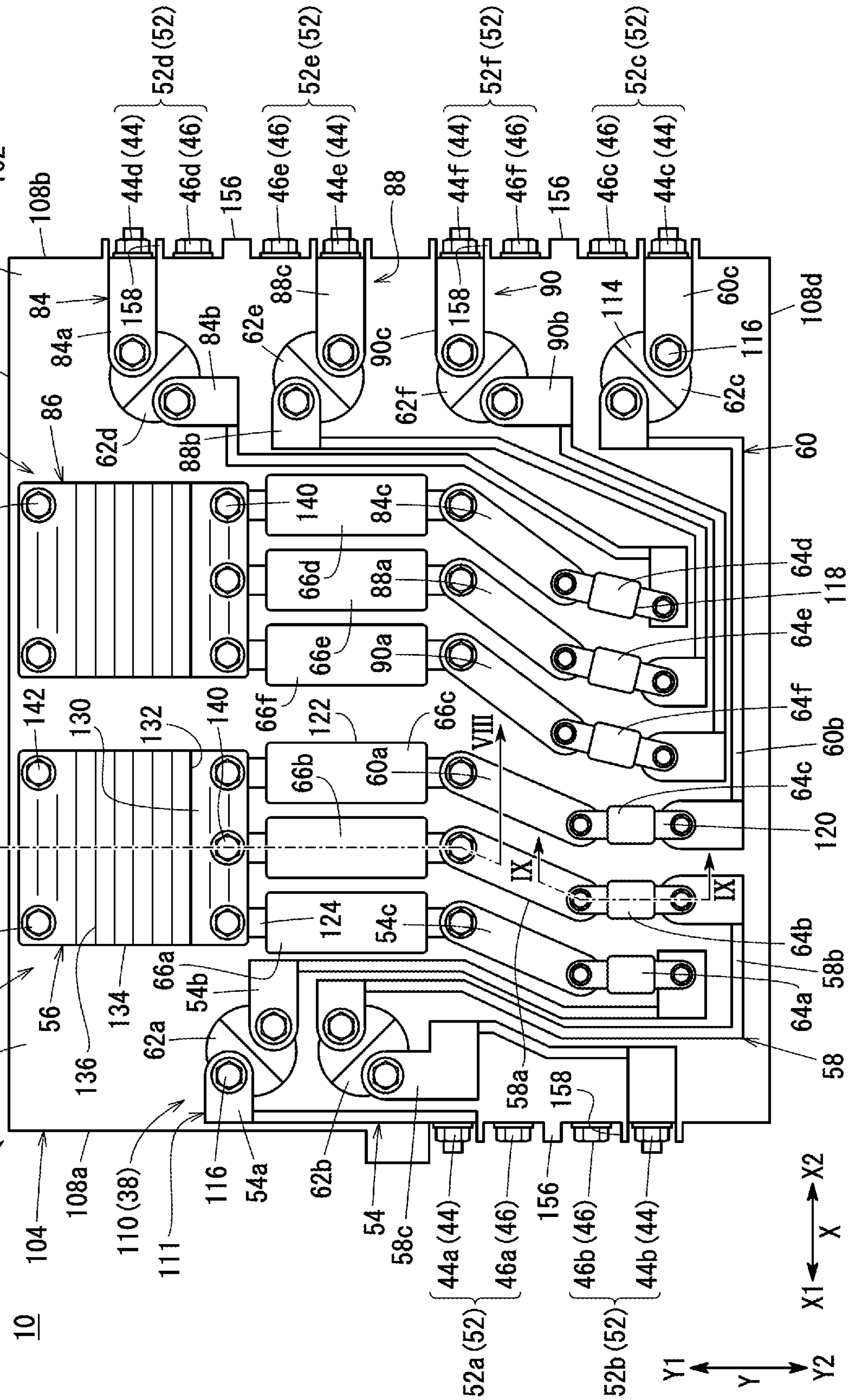
FIG. 4
10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102
108b
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88
108c
84
84a
158
84b
62e
88c
90c
90
90b
60c
114
116
108d
62f
62c
80
86
62d
88b
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142
140
84c
66d
88a
64d
118
66e
90a
64e
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132
66f
122
66c
64f
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66b
60a
64c
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IX
64b
124
54c
58b
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66a
54b
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64a
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58a
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108a
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116
54a
62b
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58c
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46a (46)
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44b (44)
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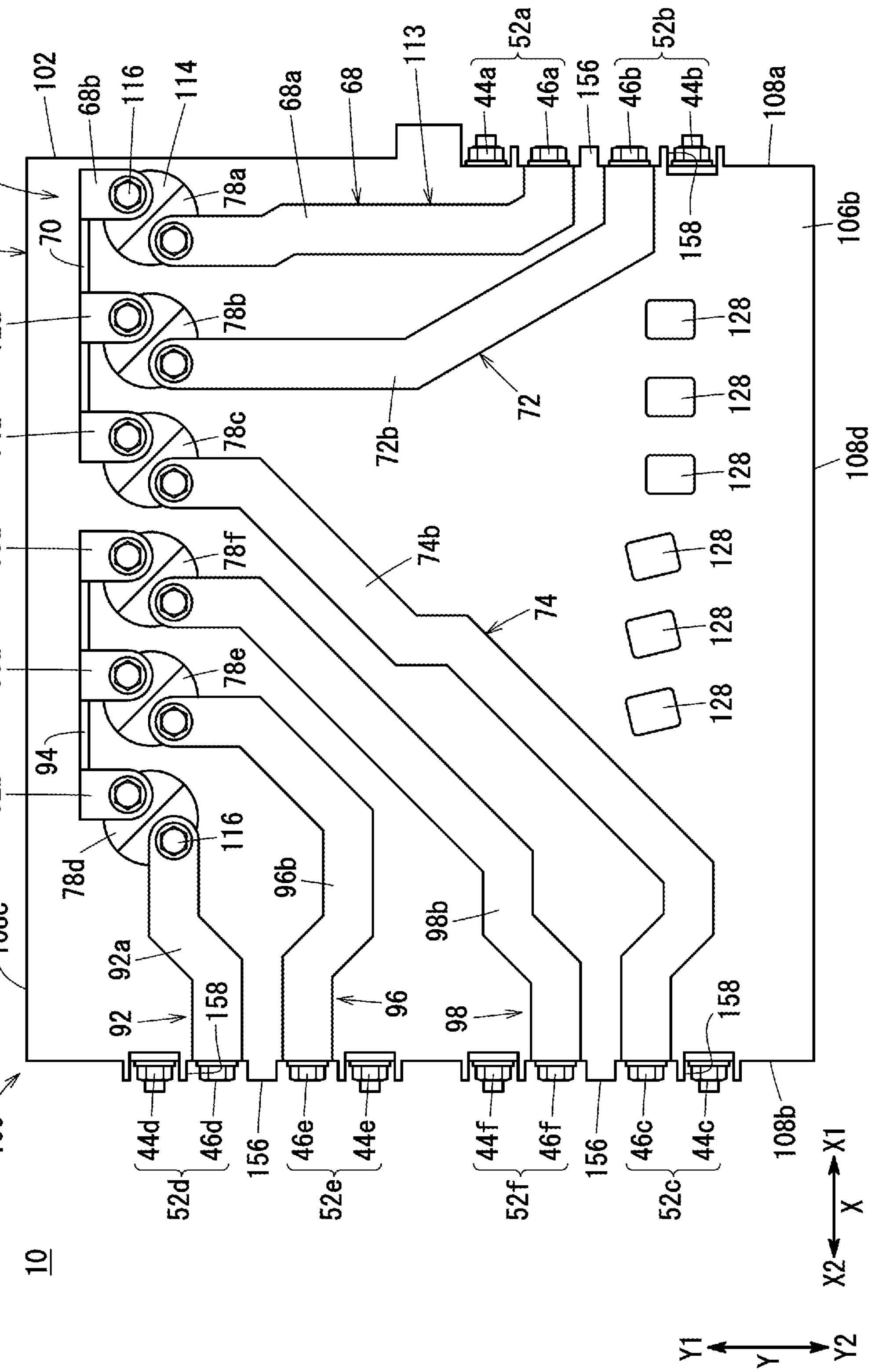
52b (52)
X1
X
X2
Y1
Y
Y2

FIG. 5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104
112 (38)
68b
116
114
68a
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113
44a
46a
52a
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108a
78a
70
158
106b
72a
78b
128
72b
72
74a
78c
108d
74b
98a
78f
74
96a
78e
94
92b
78d
108c
96b
98b
92a
92
96
98
44d
46d
46e
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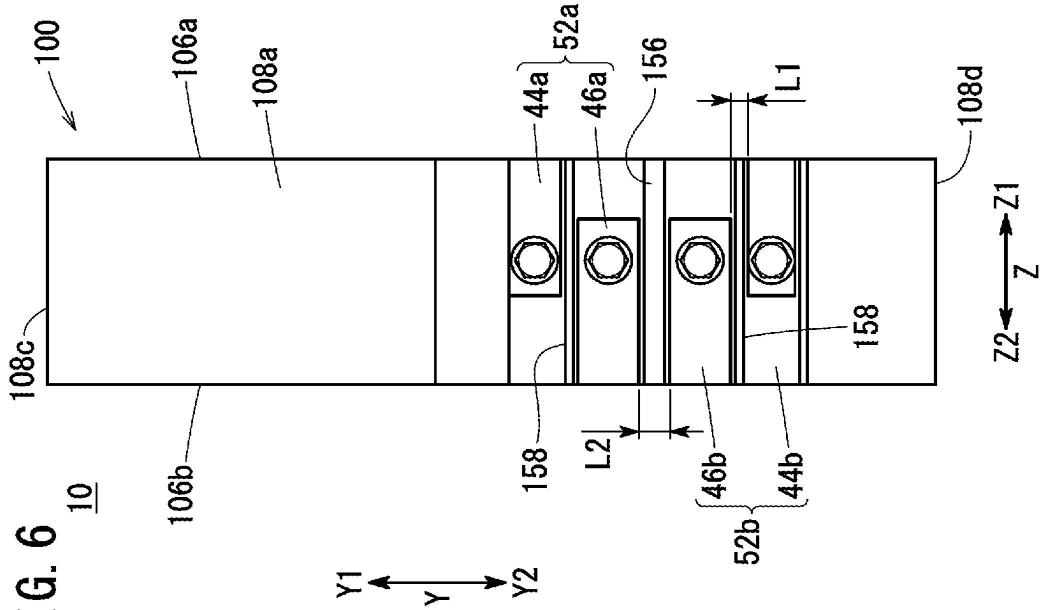
X2
Y1
Y
Y2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46a
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L1
108d
Z1
Z
Z2
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108c
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L2
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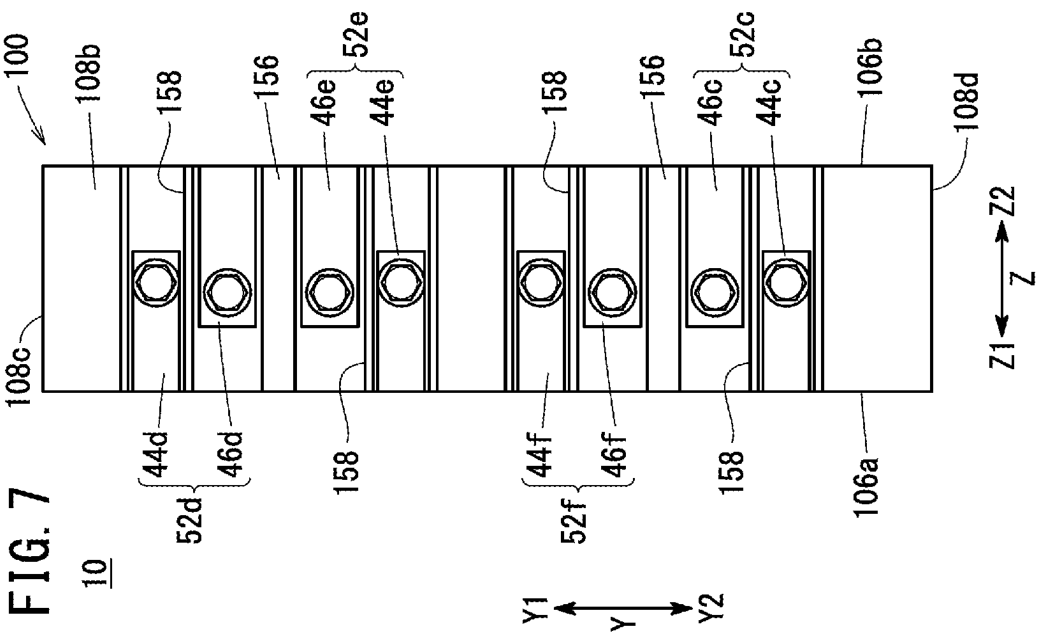
44b
52b
Y1
Y
Y2

FIG. 6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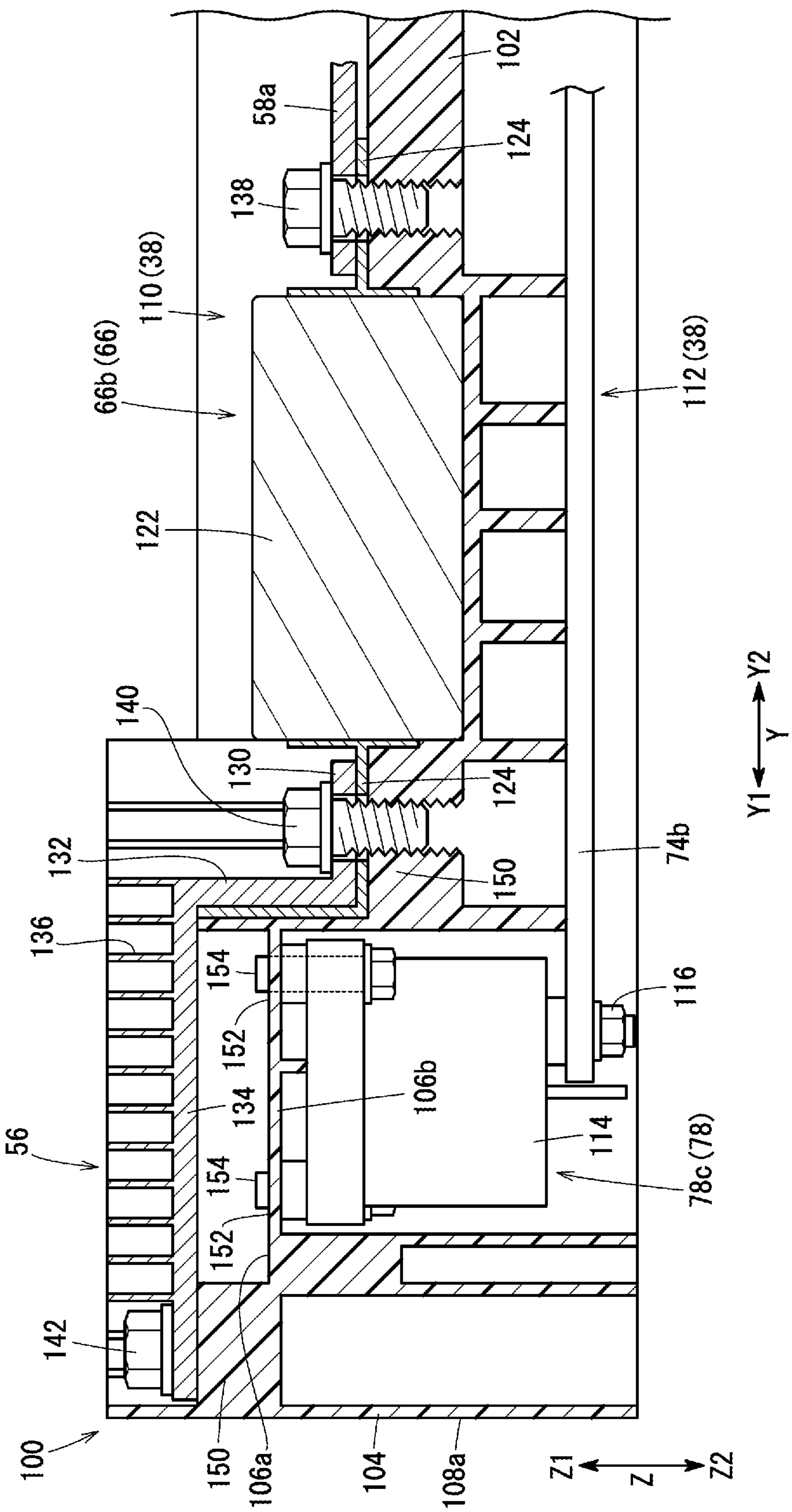
46c
52c
44c
106b
108d
Z2
Z
Z1
108c
44d
46d
52d
158
44f
46f
52f
158
106a
Y1
Y
Y2

FIG. 7

10

FIG. 8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58a
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138
110 (38)
66b (66)
122
112 (38)
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154
152
116
106b
134
114
78c (78)
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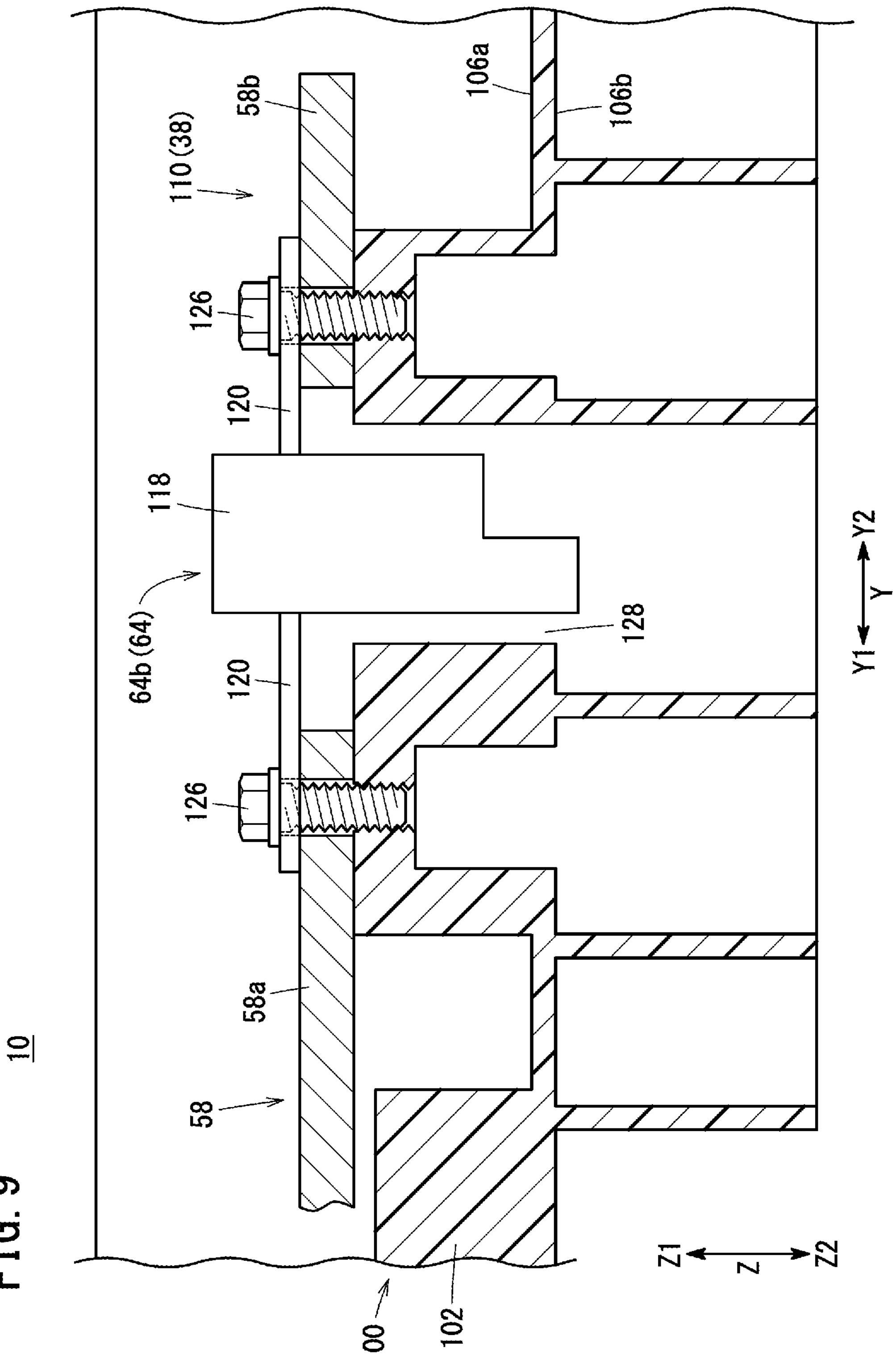
152
142
150
106a
104
108a
Y1
Y
Y2
Z1
Z
Z2

FIG. 9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106b
110 (38)
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64b (64)
128
Y2
Y
Y1
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126
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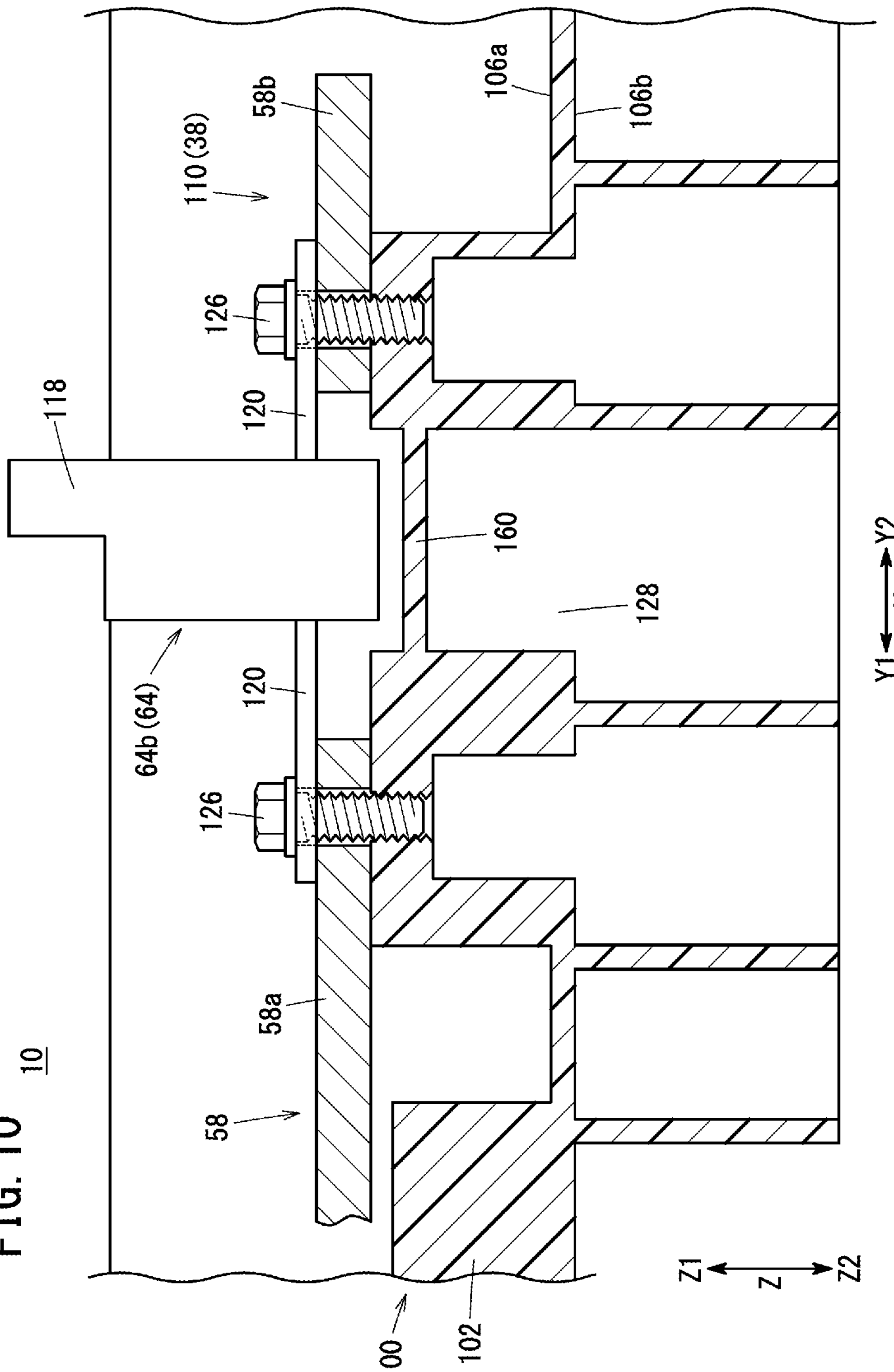
Z1
Z
Z2
100
102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58b
100
102
106a
106b
110 (38)
118
120
120
126
126
128
160
64b (64)
Y1
Y
Y2
Z1
Z
Z2

FIG. 10

ELECTRIC POWER DEVICE AND MOVING OBJECT WITH ELECTRIC POWER DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2023-046185 filed on Mar. 23, 2023, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to an electric power device and a moving object with the electric power device.

Description of the Related Art

In recent years, efforts toward realization of low-carbon or decarbonized society have been activated, and research and development have been conducted on electric power devices for reducing CO_2 emission and improving energy efficiencies also in moving objects including aircrafts.

For example, JP 3765229 B2 discloses a junction box (electric power device) including a direct current circuit having a bus bar.

SUMMARY OF THE INVENTION

To provide an electric power device capable of reliably insulating a positive electrode and a negative electrode by a simple structure.

An object of the present invention is to solve the above-described problems.

One aspect of the present invention is to provide an electric power device including a direct current circuit and a support board having electrical insulation properties and configured to support the direct current circuit, where in the direct current circuit including a positive connection terminal and a negative connection terminal, each configured to be connected to an external circuit; a first circuit portion including a first bus bar electrically connected to one of the positive connection terminal or the negative connection terminal; and a second circuit portion including a second bus bar electrically connected to another of the positive connection terminal or the negative connection terminal, the first circuit portion disposed on a first surface of the support board and the second circuit portion disposed on a second surface of the support board are electrically separated from each other by the support board, and a set of the positive connection terminal and the negative connection terminal which are paired is arranged on a same side surface of the support board.

Another aspect of the present invention is a moving object including the electric power device described above.

According to the present invention, the positive electrode and the negative electrode can be reliably insulated.

The above and other objects features and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings in which a preferred embodiment of the present invention is shown by way of illustrative example.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic view of an aircraft according to an embodiment of the present invention;

FIG. **2** is a block diagram showing a power generation module and an electric power device;

FIG. **3** is a circuit diagram of the electric power device;

FIG. **4** is a front view of the electric power device;

FIG. **5** is a rear view of the electric power device;

FIG. **6** is a side view of the electric power device;

FIG. **7** is a side view of the electric power device;

FIG. **8** is a cross-sectional view taken along line VIII-VIII of FIG. **4**;

FIG. **9** is a cross-sectional view taken along line IX-IX of FIG. **4**; and

FIG. **10** is a cross-sectional view showing an electric power device according to a modification.

DETAILED DESCRIPTION OF THE INVENTION

An electric power device **10** and a moving object **12** equipped with the electric power device **10** according to an embodiment of the present invention will be described below with reference to the drawings. As shown in FIG. **1**, the electric power device **10** according to the present embodiment is mounted on, for example, an aircraft **14** as the moving object **12**. The aircraft **14** is an electric vertical take-off and landing (eVTOL) aircraft.

The aircraft **14** includes a fuselage **16**, eight VTOL rotors **18**, and two cruise rotors **20**. The VTOL rotors **18** generate a thrust in the vertical direction on the aircraft **14**. One VTOL motor **22** is connected to each VTOL rotor **18**. The VTOL rotor **18** is rotated by the VTOL motor **22**. The cruise rotors **20** generate a thrust in the horizontal direction on the aircraft **14**. Two cruise motors **24** are connected to each cruise rotor **20**. The cruise rotor **20** is rotated by the cruise motor **24**. The number of VTOL rotors **18**, VTOL motors **22**, cruise rotors **20**, and cruise motors **24** can be set as desired.

A power generation module **26** and the electric power device **10** are disposed inside the fuselage **16**. As shown in FIG. **2**, the power generation module **26** includes an engine **30**, a generator **32**, and a power control unit (PCU) **34**. The engine **30** is, for example, a gas turbine engine, but is not limited thereto. The generator **32** generates power by being driven by the engine **30**. The PCU **34** converts the three phase power generated by the power generator **32** into DC power.

The electric power device **10** is a junction box for supplying DC power supplied from the PCU **34** to an electric load **36**. The electric load **36** includes various electronic devices such as the VTOL motors **22**, the cruise motors **24**, and inverters. The moving object **12** may not be limited to an aircraft and may be, for example, a vehicle, a ship, or the like. The electric power device **10** is not necessarily mounted on the moving object **12**.

As shown in FIG. **3**, the electric power device **10** includes a direct current circuit **38**. The direct current circuit **38** includes a first electric circuit **40** and a second electric circuit **42**. The first electric circuit **40** includes a plurality of positive connection terminals **44**, a plurality of negative connection terminals **46**, a first positive circuit portion **48**, and a first negative circuit portion **50**. The positive connection terminal **44** and the negative connection terminal **46** of the first electric circuit **40** are electrically connected to an external circuit.

The first electric circuit **40** includes a first positive connection terminal **44***a*, a second positive connection terminal **44***b*, and a third positive connection terminal **44***c* as the plurality of positive connection terminals **44**. The first electric circuit **40** includes a first negative connection terminal **46**a, a second negative connection terminal **46**b, and a third negative connection terminal **46**c as the plurality of negative connection terminals **46**.

The first positive connection terminal **44**a and the first negative connection terminal **46**a form a first set of terminals **52**a. The first set of terminals **52**a is electrically connected to the PCU **34**. The second positive connection terminal **44**b and the second negative connection terminal **46**b form a second set of terminals **52**b. The electric power output via the second set of terminals **52**b is supplied to, for example, two of the VTOL motors **22** and one of the cruise motors **24**. The third positive connection terminal **44**c and the third negative connection terminal **46**c form a third set of terminals **52**c. The electric power output via the third set of terminals **52**c is supplied to, for example, two of the VTOL motors **22** and one of the cruise motors **24**.

The first positive circuit portion **48** electrically connects the first positive connection terminal **44**a, the second positive connection terminal **44**b, and the third positive connection terminal **44**c to each other. The first positive circuit portion **48** includes a first positive conductor **54**, a first junction **56**, a second positive conductor **58**, and a third positive conductor **60**.

The first positive conductor **54** electrically connects the first positive connection terminal **44**a and the first junction **56** to each other. The second positive conductor **58** electrically connects the first junction **56** and the second positive connection terminal **44**b to each other. The third positive conductor **60** electrically connects the first junction **56** and the third positive connection terminal **44**c to each other. The first to third positive conductors **54**, **58**, **60** are each provided with a positive contactor **62**, an ammeter **64**, and a fuse **66**.

The first negative circuit portion **50** electrically connects the first negative connection terminal **46**a, the second negative connection terminal **46**b, and the third negative connection terminal **46**c to each other. The first negative circuit portion **50** includes a first negative conductor **68**, a first junction **70**, a second negative conductor **72**, and a third negative conductor **74**.

The first negative conductor **68** electrically connects the first negative connection terminal **46**a and the first junction **70** to each other. The second negative conductor **72** electrically connects the first junction **70** and the second negative connection terminal **46**b to each other. The third negative conductor **74** electrically connects the first junction **70** and the third negative connection terminal **46**c to each other. The first to third negative conductors **68**, **72**, **74** are provided with one negative contactor **78** each.

The second electric circuit **42** includes a plurality of positive connection terminals **44**, a plurality of negative connection terminals **46**, a second positive circuit portion **80**, and a second negative circuit portion **82**. The positive connection terminal **44** and the negative connection terminal **46** of the second electric circuit **42** are electrically connected to an external circuit.

The second electric circuit **42** includes a fourth positive connection terminal **44**d, a fifth positive connection terminal **44**e, and a sixth positive connection terminal **44**f as the plurality of positive connection terminals **44**. The second electric circuit **42** includes a fourth negative connection terminal **46**d, a fifth negative connection terminal **46**e, and a sixth negative connection terminal **46**f as the plurality of negative connection terminals **46**.

The fourth positive connection terminal **44**d and the fourth negative connection terminal **46**d form a fourth set of terminals **52**d. The fourth set of terminals **52**d is electrically connected to the PCU **34**. The fifth positive connection terminal **44**e and the fifth negative connection terminal **46**e form a fifth set of terminals **52**e. The electric power output via the fifth set of terminals **52**e is supplied to, for example, two of the VTOL motors **22** and one of the cruise motors **24**. The sixth positive connection terminal **44**f and the sixth negative connection terminal **46**f form a sixth set of terminals **52**f. The electric power output via the sixth set of terminals **52**f is supplied to, for example, two of the VTOL motors **22** and one of the cruise motors **24**.

The second positive circuit portion **80** electrically connects the fourth positive connection terminal **44**d, the fifth positive connection terminal **44**e, and the sixth positive connection terminal **44**f to each other. The second positive circuit portion **80** includes a fourth positive conductor **84**, a second junction **86**, a fifth positive conductor **88** and a sixth positive conductor **90**.

The fourth positive conductor **84** electrically connects the fourth positive connection terminal **44**d and the second junction **86** to each other. The fifth positive conductor **88** electrically connects the second junction **86** and the fifth positive connection terminal **44**e to each other. The sixth positive conductor **90** electrically connects the second junction **86** and the sixth positive connection terminal **44**f to each other. The fourth to sixth positive conductors **84**, **88**, **90** are each provided with the positive contactor **62**, an ammeter **64**, and a fuse **66**.

The second negative circuit portion **82** electrically connects the fourth negative connection terminal **46**d, the fifth negative connection terminal **46**e, and the sixth negative connection terminal **46**f to each other. The second negative circuit portion **82** includes a fourth negative conductor **92**, a second junction **94**, a fifth positive conductor **96** and a sixth negative conductor **98**.

The fourth negative conductor **92** electrically connects the fourth negative connection terminal **46**d and the second junction **94** to each other. The fifth negative conductor **96** electrically connects the second junction **94** and the fifth negative connection terminal **46**e to each other. The sixth negative conductor **98** electrically connects the second junction **94** and the sixth negative connection terminal **46**f to each other. The fourth to sixth negative conductors **92**, **96**, **98** are provided with one negative contactor **78** each.

As shown in FIGS. **4** to **7**, the electric power device **10** includes a support board **100** that supports the direct current circuit **38** and has electrical insulation properties. The support board **100** includes a plate-shaped support main body **102** and a peripheral wall **104** (see FIG. **8**). The support main body **102** is formed in a quadrangular shape in a front view shown in FIG. **4**.

As shown in FIGS. **4** to **9**, the support main body **102** has a first surface **106**a as one surface and a second surface **106**b as the other surface. The first surface **106**a faces in the arrow Z1 direction, and the second surface **106**b faces in the arrow Z2 direction.

As shown in FIGS. **4** and **5**, the peripheral wall **104** surrounds the outer periphery of the support main body **102**. The peripheral wall **104** protrudes from the support main body **102** in the arrow Z1 direction and the arrow **22** direction (see FIG. **8**). The peripheral wall **104** has a first side surface **108**a, a second side surface **108**b, a third side surface **108**c, and a fourth side surface **108**d. The first side surface **108**a faces in the arrow X1 direction, and the second side surface **108**b faces in the arrow X2 direction. That is, the first side surface **108**a and the second side surface **108**b face in opposite directions. The third side surface **108**c faces in the arrow Y1 direction, and the fourth side surface **108**d faces in the arrow Y2 direction.

As shown in FIG. **4**, a first circuit portion **110** including the first positive circuit portion **48** and the second positive circuit portion **80** are disposed on the first surface **106***a* of the support board **100**. The first circuit portion **110** includes a plurality of first bus bars **111**.

As shown in FIG. **5**, a second circuit portion **112** including the first negative circuit portion **50** and the second negative circuit portion **82** is disposed on the second surface **106***b* of the support board **100**. The second circuit portion **112** includes a plurality of second bus bars **113**.

As shown in FIG. **6**, the first side surface **108***a* of the support board **100** is provided with the first set of terminals **52***a* and the second set of terminals **52***b*. As shown in FIG. **7**, the third set of terminals **52***c*, the fourth set of terminals **52***d*, the fifth set of terminals **52***e*, and the sixth set of terminals **52***f* are disposed on the second side surface **108***b* of the support board **100**. As shown in FIGS. **6** and **7**, the positive connection terminals **44** and the second surface **106***b* of the support board **100** are separated from each other. The negative connection terminals **46** and the first surface **106***a* of the support board **100** are separated from each other.

As shown in FIG. **4**, only the first circuit portion **110** is arranged on the first surface **106***a* of the support board **100**, and the second circuit portion **112** is not arranged thereon. As shown in FIG. **5**, only the second circuit portion **112** is arranged on the second surface **106***b* of the support board **100**, and the first circuit portion **110** is not arranged thereon. That is, the first circuit portion **110** and the second circuit portion **112** are electrically separated by the support board **100** having electrical insulation properties. The support board **100** has electrical insulation properties and a certain thickness. Therefore, the first circuit portion **110** and the second circuit portion **112** can be reliably insulated from each other by the support board **100**.

Hereinafter, the arrangement structure of the direct current circuit **38** with respect to the support board **100** will be described in detail. As shown in FIG. **4**, the first circuit portion **110** includes six positive contactors **62**, six ammeters **64**, and six fuses **66**. As shown in FIG. **5**, the second circuit portion **112** includes six negative contactors **78**.

As shown in FIG. **4**, the six positive contactors **62** connect or disconnect the first to sixth positive conductors **54**, **58**, **60**, **84**, **88**, **90**. As shown in FIG. **5**, the six negative contactors **78** connect or disconnect the first to sixth negative conductors **68**, **72**, **74**, **92**, **96**, **98**. As shown in FIGS. **5** and **8**, the negative contactor **78** includes a contactor body **114** and a pair of contact points **116**. The contactor body **114** is formed in a columnar shape, for example. The pair of contact points **116** are provided, for example, on an end surface of the contactor body **114**. The positive contactor **62** has the same configuration as the negative contactor **78**.

As shown in FIG. **4**, the six ammeters **64** measure currents flowing through the first to sixth positive conductors **54**, **58**, **60**, **84**, **88**, **90**. As shown in FIG. **9**, the ammeter **64** includes a base portion **118** and a pair of contact points **120**. The base portion **118** extends in one direction (the arrow Z direction). The base portion **118** is formed in a rectangular parallelepiped shape, for example. The pair of contact points **120** protrude from the base portion **118** in opposite directions (the arrow Y direction). The pair of contact points **120** are provided at one end portion of the base portion **118** in the extending direction.

As shown in FIG. **4**, when an excessive current over the upper limit value flows through any one of the first to sixth positive conductors **54**, **58**, **60**, **84**, **88**, **90**, corresponding one of the six fuses **66** interrupts the current flowing through the one of the first to sixth positive conductors **54**, **58**, **60**, **84**, **88**, **90**. As shown in FIG. **8**, the fuse **66** includes a fuse body **122** and a pair of contact points **124**. The fuse body **122** extends in one direction (the arrow Y direction). The pair of contact points **124** are provided at both ends of the fuse body **122** in the longitudinal direction.

Hereinafter, the positive contactors **62** provided in the first to sixth positive conductors **54**, **58**, **60**, **84**, **88**, **90** may be referred to as first to sixth positive contactors **62***a* to **62***f*. The negative contactors **78** provided in the first to sixth negative conductors **68**, **72**, **74**, **92**, **96**, **98** may be referred to as first to sixth negative contactors **78***a* to **78***f*.

As shown in FIG. **4**, the first to sixth positive contactors **62***a* to **62***f*, the six ammeters **64**, and the six fuses **66** are attached to the first surface **106***a* of the support board **100**. The first to sixth positive contactors **62***a* to **62***f* are fixed to the first surface **106***a* of the support board **100** by bolts (not shown).

The first positive contactor **62***a* and the second positive contactor **62***b* are arranged along the first side surface **108***a* of the support board **100**. The first positive contactor **62***a* is positioned in the direction of the arrow Y1 with respect to the second positive contactor **62***b*. The third positive contactor **62***c*, the fourth positive contactor **62***d*, the fifth positive contactor **62***e*, and the sixth positive contactor **62***f* are disposed along the second side surface **108***b* of the support board **100**. The fourth positive contactor **62***d*, the fifth positive contactor **62***e*, the sixth positive contactor **62***f*, and the third positive contactor **62***c* are arranged in this order in the arrow Y2 direction.

The six ammeters **64** are arranged along the fourth side surface **108***d*. In other words, the six ammeters **64** are arranged along the arrow X direction. The ammeters **64** are fixed to the support board **100** by bolts **126** (see FIG. **9**). The six fuses **66** are arranged in the arrow X direction in the central portion of the first surface **106***a* of the support board **100**. The fuses **66** are fixed to the support board **100** by bolts **138**, **140** (see FIG. **8**).

Hereinafter, the ammeters **64** provided in the first to sixth positive conductors **54**, **58**, **60**, **84**, **88**, **90** may be referred to as first to sixth ammeters **64***a* to **64***f*. The fuses **66** provided in the first to sixth positive conductors **54**, **58**, **60**, **84**, **88**, **90** may be referred to as first to sixth fuses **66***a* to **66***f*.

As shown in FIGS. **4** and **6**, the first positive connection terminal **44***a* is disposed on the first side surface **108***a*. The first positive connection terminal **44***a* extends from the middle portion of the first side surface **108***a* in the arrow Z direction to the first surface **106***a* of the support board **100**.

As shown in FIG. **4**, a first positive conductor **54** is electrically connected to the first positive connection terminal **44***a*. The first positive conductor **54** includes three first positive bus bars **54***a* to **54***c* as first bus bars **111**. The first positive bus bar **54***an* electrically connects the first positive connection terminal **44***a* and one of the contact points **116** of the first positive contactor **62***a* to each other. The first positive bus bar **54***a* extends from the first positive connection terminal **44***a* in the arrow Y1 direction.

The first positive bus bar **54***b* electrically connects the other of the contact points **116** of the first positive contactor **62***a* and the first ammeter **64***a* to each other. The first positive bus bar **54***b* extends from the first positive contactor **62***a* in the arrow Y2 direction.

The first positive bus bar **54***c* electrically connects the other of the contact points **120** of the first ammeter **64***a* and one of the contact points **124** of the first fuse **66***a* to each other. The first positive bus bar **54***c* extends from the first ammeter **64***a* in the arrow Y1 direction.

Each of the contact points **120** of the first ammeter **64***a* and the first positive bus bars **54***b*, **54***c* are fixed to the support board **100** by bolts **126**. As shown in FIG. **9**, the base portion **118** of the ammeter **64** is inserted into a through hole **128** formed in the support board **100**. The other end of the base portion **118** of the ammeter **64** is positioned on the second surface **106***b* side of the support board **100**.

As shown in FIG. **4**, the first junction **56** is electrically connected to the other of the contact points **124** of the first fuse **66***a*. The first junction **56** is positioned at an end portion in the direction of the arrow Y1. The first junction **56** is formed in a plate shape. The first junction **56** is formed in a quadrangular shape in the front view shown in FIG. **4**.

As shown in FIG. **8**, the first junction **56** includes a junction body **130**, a first heat dissipation plate portion **132**, a second heat dissipation plate portion **134**, and a plurality of fins **136**. The junction body **130** is positioned at an end portion of the first junction **56** in the arrow Y2 direction.

As shown in FIG. **4**, the other of the contact points **124** of the first fuse **66***a*, one of the contact points **124** of the second fuse **66***b*, and one of the contact points **124** of the third fuse **66***c* are electrically connected to the junction body **130** of the first junction **56**. The junction body **130** of the first junction **56** and the other of the contact points **124** of the first fuse **66***a* are fixed to the support board **100** by a bolt **140**. The same applies to the second fuse **66***b* and the third fuse **66***c*. The three bolts **140** are arranged at intervals in the arrow X direction. Two of the three bolts **140** are positioned at the corners of the first junction **56**.

As shown in FIG. **8**, the first heat dissipation plate portion **132** protrudes from the junction body **130** in the arrow Z1 direction. The second heat dissipation plate portion **134** protrudes in the arrow Y1 direction from the arrow Z1 side end of the first heat dissipation plate portion **132**. As shown in FIGS. **4** and **8**, the arrow Y1 side end portion of the second heat dissipation plate portion **134** is fixed to the support board **100** by two bolts **142**. The two bolts **142** are positioned at the corners of the first junction **56**. The number and positions of the bolts **142** for fixing the second heat dissipation plate portion **134** to the support board **100** are appropriately set. The plurality of fins **136** protrude from the junction body **130** and the second heat dissipation plate portion **134** in the arrow Z1 direction. The first heat dissipation plate portion **132** is not provided with the fins **136**. The plurality of fins **136** are arranged at intervals in the arrow Y direction. The fins **136** extend in the arrow X direction (see FIG. **4**).

As shown in FIG. **4**, the second fuse **66***b* is provided in the second positive conductor **58**. The second fuse **66***b* is positioned in the arrow X2 direction with respect to the first fuse **66***a*. The second positive conductor **58** includes three second positive bus bars **58***a* to **58***c* as the first bus bars **111**.

The second positive bus bar **58***an* electrically connects the other of the contact points **124** of the second fuse **66***b* and one of the contact points **120** of the second ammeter **64***b* to each other. The second ammeter **64***b* is positioned in the arrow X2 direction with respect to the first ammeter **64***a*. The second positive bus bar **58***a* is disposed in parallel to the first positive bus bar **54***c*. The second positive bus bar **58***a* extends from the second fuse **66***b* in the arrow Y2 direction. The second positive bus bar **58***a* is positioned in the arrow X2 direction with respect to the first positive bus bar **54***c*.

The second positive bus bar **58***b* electrically connects the other of the contact points **120** of the second ammeter **64***b* and one of the contact points **116** of the second positive contactor **62***b* to each other. The second positive contactor **62***b* is positioned in the arrow Y2 direction with respect to the first positive contactor **62***a*. The second positive bus bar **58***b* extends from the second ammeter **64***b* in the arrow X1 direction, and is then bent in the arrow Y1 direction.

The second positive bus bar **58***c* electrically connects the other of the contact points **116** of the second positive contactor **62***b* and the second positive connection terminal **44***b* to each other. As shown in FIG. **6**, the second positive connection terminal **44***b* is disposed on the first side surface **108***a*. The second positive connection terminal **44***b* is positioned in the arrow Y2 direction with respect to the first positive connection terminal **44***a*.

As shown in FIG. **4**, the third fuse **66***c* is provided in the third positive conductor **60**. The third fuse **66***c* is positioned in the arrow X2 direction with respect to the second fuse **66***b*. The third positive conductor **60** includes three third positive bus bars **60***a* to **60***c* as the first bus bars **111**.

The third positive bus bar **60***an* electrically connects the other of the contact points **124** of the third fuse **66***c* and one of the contact points **120** of the third ammeter **64***c* to each other. The third ammeter **64***c* is positioned in the arrow X2 direction with respect to the second ammeter **64***b*. The third positive bus bar **60***a* is disposed in parallel to the second positive bus bar **58***a*. The third positive bus bar **60***a* extends from the third fuse **66***c* in the arrow Y2 direction. The third positive bus bar **60***a* is positioned in the arrow X2 direction with respect to the second positive bus bar **58***a*.

The third positive bus bar **60***b* electrically connects the other of the contact points **120** of the third ammeter **64***c* and one of the contact points **116** of the third positive contactor **62***c* to each other. The third positive contactor **62***c* is positioned at a corner of the support board **100** in the arrows X2 and Y2 directions. The third positive bus bar **60***b* extends from the third ammeter **64***c* in the arrow X2 direction.

The third positive bus bar **60***c* electrically connects the other of the contact points **116** of the third positive contactor **62***c* and the third positive connection terminal **44***c* to each other. As shown in FIG. **7**, the third positive connection terminal **44***c* is disposed on the second side surface **108***b* of the support board **100**. The third positive connection terminal **44***c* is positioned at the end portion of the second side **108***b* in the arrow Y2 direction.

The fourth positive connection terminal **44***d* is arranged at the end portion of the second side **108***b* in the arrow Y1 direction. The fourth positive connection terminal **44***d* extends from the middle portion of the third side surface **108***c* in the arrow Z direction to the first surface **106***a*. The fourth positive conductor **84** is electrically connected to the fourth positive connection terminal **44***d*. The fourth positive conductor **84** includes three fourth positive bus bars **84***a* to **84***c* as the first bus bars **111**.

As shown in FIG. **4**, the fourth positive bus bar **84***an* electrically connects the fourth positive connection terminal **44***d* to one of the contact points **116** of the fourth positive contactor **62***d*. The fourth positive contactor **62***d* is positioned in the arrow Y1 direction with respect to the third positive contactor **62***c*. The fourth positive bus bar **84***a* extends from the fourth positive connection terminal **44***d* in the arrow X1 direction.

The fourth positive bus bar **84***b* electrically connects the other of the contact points **116** of the fourth positive contactor **62***d* and one of the contact points **120** of the fourth ammeter **64***d* to each other. The fourth ammeter **64***d* is positioned in the arrow X2 direction with respect to the third ammeter **64***c*. The fourth positive bus bar **84***b* extends from the fourth positive contactor **62***d* in the arrow Y2 direction.

The fourth positive bus bar **84***c* electrically connects the other of the contact points **120** of the fourth ammeter **64***d* and one of the contact points **124** of the fourth fuse **66***d* to each other. The fourth fuse **66***d* is positioned in the arrow Y1 direction with respect to the fourth ammeter **64***d*. The fourth positive bus bar **84***c* extends from the fourth ammeter **64***d* in the arrow Y1 direction. The other of the contact points **124** of the fourth fuse **66***d* is electrically connected to the second junction **86**.

The second junction **86** has the same configuration as the first junction **56**. Therefore, detailed descriptions of the configuration of the second junction **86** will be omitted. The second junction **86** is arranged with a gap from the first junction **56** in the arrow X2 direction. The first junction **56** and the second junction **86** are electrically separated from each other.

The other of the contact points **124** of the fourth fuse **66***d*, one of the contact points **124** of the fifth fuse **66***e*, and one of the contact points **124** of the sixth fuse **66***f* are electrically connected to the junction body **130** of the second junction **86**. The second junction **86** and the other of the contact points **124** of the fourth fuse **66***d* are fixed to the support board **100** by a bolt **140**. The same applies to the fifth fuse **66***e* and the sixth fuse **66***f*. The fifth fuse **66***e* is positioned in the arrow X1 direction with respect to the fourth fuse **66***d*. The sixth fuse **66***f* is positioned in the arrow X1 direction with respect to the fifth fuse **66***e*.

The fifth fuse **66***e* is provided in the fifth positive conductor **88**. The fifth positive conductor **88** includes three fifth positive bus bars **88***a* to **88***c* as the first bus bars **111**. The fifth positive bus bar **88***an* electrically connects the other of the contact points **124** of the fifth fuse **66***e* and one of the contact points **120** of the fifth ammeter **64***e* to each other. The fifth ammeter **64***e* is positioned in the arrow X1 direction with respect to the fourth ammeter **64***d*. The fifth positive bus bar **88***a* is disposed in parallel to the fourth positive bus bar **84***c*. The fifth positive bus bar **88***a* is positioned in the arrow X1 direction with respect to the fourth positive bus bar **84***c*.

The fifth positive bus bar **88***b* electrically connects the other of the contact points **120** of the fifth ammeter **64***e* and one of the contact points **116** of the fifth positive contactor **62***e* to each other. The fifth positive contactor **62***e* is positioned in the arrow Y2 direction with respect to the fourth positive contactor **62***d*. The fifth positive bus bar **88***b* extends from the fifth ammeter **64***e* in the arrow X2 direction, and is then bent in the arrow Y1 direction.

The fifth positive bus bar **88***c* electrically connects the other of the contact points **116** of the fifth positive contactor **62***e* and the fifth positive connection terminal **44***e* to each other. The fifth positive connection terminal **44***e* is disposed on the second side surface **108***b* of the support board **100**. The fifth positive connection terminal **44***e* is positioned between the third positive connection terminal **44***c* and the fourth positive connection terminal **44***d* (see FIG. **7**).

The sixth fuse **66***f* is provided in the sixth positive conductor **90**. The sixth positive conductor **90** includes three sixth positive bus bars **90***a* to **90***c* as the first bus bars **111**. The sixth positive bus bar **90***an* electrically connects the other of the contact points **124** of the sixth fuse **66***f* and one of the contact points **120** of the sixth ammeter **64***f* to each other. The sixth positive bus bar **90***a* is disposed in parallel to the fifth positive bus bar **88***a*. The sixth ammeter **64***f* is positioned in the arrow X1 direction with respect to the fifth ammeter **64***e*.

The sixth positive bus bar **90***b* electrically connects the other of the contact points **120** of the sixth ammeter **64***f* and one of the contact points **116** of the sixth positive contactor **62***f* to each other. The sixth positive contactor **62***f* is positioned between the fifth positive contactor **62***e* and the third positive contactor **62***c*. The sixth positive bus bar **90***b* extends from the sixth ammeter **64***f* in the arrow X2 direction, and is then bent in the arrow Y1 direction.

The sixth positive bus bar **90***c* electrically connects the other of the contact points **116** of the sixth positive contactor **62***f* and the sixth positive connection terminal **44***f* to each other. The sixth positive connection terminal **44***f* is disposed on the second side surface **108***b* of the support board **100**. The sixth positive connection terminal **44***f* is positioned between the third positive connection terminal **44***c* and the fifth positive connection terminal **44***e* (see FIG. **7**).

As shown in FIGS. **5** and **6**, the first negative connection terminal **46***a* is disposed on the first side surface **108***a*. The first negative connection terminal **46***a* is adjacent to the first positive connection terminal **44***a* in the arrow Y2 direction.

As shown in FIG. **5**, a first negative conductor **68** is electrically connected to the first negative connection terminal **46***a*. The first negative conductor **68** includes two first negative bus bars **68***a*, **68***b* as the second bus bars **113**. The first negative bus bar **68***an* electrically connects the first negative connection terminal **46***a* and one of the contact points **116** of the first negative contactor **78***a* to each other. The first negative bus bar **68***a* extends from the first negative connection terminal **46***a* in the arrow Y1 direction. The first negative contactor **78***a* is positioned at an end portion of the support board **100** in the arrow X1 direction.

The first negative bus bar **68***b* electrically connects the other of the contact points **116** of the first negative contactor **78***a* and the first junction **70** to each other. The first negative bus bar **68***b* extends from the first negative contactor **78***a* in the arrow Y1 direction.

The first junction **70** is a plate-shaped portion extending in the arrow X direction. A second negative conductor **72** and a third negative conductor **74** are electrically connected to the first junction **70**. The second negative conductor **72** includes two second negative bus bars **72***a*, **72***b* as the second bus bars **113**.

The second negative bus bar **72***an* electrically connects the first junction **70** and one of the contact points **116** of the second negative contactor **78***b* to each other. The second negative bus bar **72***a* is positioned in the arrow X2 direction with respect to the first negative bus bar **68***b*. The second negative bus bar **72***a* extends from the first junction **70** in the arrow Y2 direction. The second negative contactor **78***b* is positioned in the arrow X2 direction with respect to the first negative contactor **78***a*.

The second negative bus bar **72***b* electrically connects the other of the contact points **116** of the second negative contactor **78***b* and the second negative connection terminal **46***b* to each other. The second negative bus bar **72***b* extends from the second negative contactor **78***b* in the arrow Y2 direction, and is then bent in the arrow X1 direction. As shown in FIG. **6**, the second negative connection terminal **46***b* is disposed on the first side surface **108***a* of the support board **100**. The second negative connection terminal **46***b* is adjacent to the second positive connection terminal **44***b* in the arrow Y1 direction.

As shown in FIG. **5**, the third negative conductor **74** includes two third negative bus bars **74***a*, **74***b* as the second bus bars **113**. The third negative bus bar **74***an* electrically connects the first junction **70** and one of the contact points **116** of the third negative contactor **78***c*. The third negative bus bar **74***a* is positioned in the arrow X2 direction with respect to the second negative bus bar **72***a*. The third negative bus bar **74***a* extends from the first junction **70** in the arrow Y2 direction. The third negative contactor **78***c* is positioned in the arrow X2 direction with respect to the second negative contactor **78***b*.

As shown in FIG. **8**, the third negative contactor **78***c* is positioned on the back side of the second heat dissipation plate portion **134** of the first junction **56**. In other words, the third negative contactor **78***c* and the second heat dissipation plate portion **134** of the first junction **56** are disposed to face each other with the support main body **102** interposed therebetween. The support board **100** includes first fixing portions **150** to which the first junction **56** is fixed and a second fixing portions **152** to which the third negative contactor **78***c* is fixed. The bolts **140**, **142** for fixing the first junction **56** are attached to the first fixing portions **150** of the support board **100**. The bolts **154** for fixing the third negative contactor **78***c* are attached to the second fixing portions **152** of the support board **100**. The first fixing portion **150** and the second fixing portion **152** are adjacent to each other.

As shown in FIG. **5**, the third negative bus bar **74***b* electrically connects the other of the contact points **116** of the third negative contactor **78***c* and the third negative connection terminal **46***c* to each other. The third negative bus bar **74***b* extends from the third negative contactor **78***c* in the arrow Y2 direction so as to be inclined in the arrow X2 direction. Six through holes **128** into which the ammeters **64** are inserted are arranged in the arrow X direction on the second surface **106***b* of the support board **100** in the area on the arrow X1 direction side of the third negative bus bar **74***b*.

As shown in FIG. **7**, the third negative connection terminal **46***c* is disposed on the second side surface **108***b* of the support board **100**. The third negative connection terminal **46***c* is adjacent to the third positive connection terminal **44***c* in the arrow Y1 direction.

The fourth negative connection terminal **46***d* is disposed on the second side surface **108***b* of the support board **100**. The fourth negative connection terminal **46***d* is adjacent to the fourth positive connection terminal **44***d* in the arrow Y2 direction.

As shown in FIG. **5**, a fourth negative conductor **92** is electrically connected to the fourth negative connection terminal **46***d*. The fourth negative conductor **92** includes two fourth negative bus bars **92***a*, **92***b* as the second bus bars **113**. The fourth negative bus bar **92***an* electrically connects the fourth negative connection terminal **46***d* and one of the contact points **116** of the fourth negative contactor **78***d* to each other. The fourth negative bus bar **92***a* extends from the fourth negative connection terminal **46***d* in the arrow X direction. The fourth negative contactor **78***d* is positioned in the arrow X2 direction with respect to the third negative contactor **78***c*.

The fourth negative bus bar **92***b* electrically connects the other of the contact points **116** of the fourth negative contactor **78***d* and the second junction **94** to each other. The fourth negative bus bar **92***b* extends from the fourth negative contactor **78***d* in the arrow Y1 direction.

The second junction **94** is a plate-shaped portion extending in the arrow X direction. The second junction **94** is positioned in the arrow X2 direction with respect to the first junction **70**. A fifth negative conductor **96** and a sixth negative conductor **98** are electrically connected to the second junction **94**. The fifth negative conductor **96** includes two fifth negative bus bars **96***a*, **96***b* as the second bus bars **113**.

The fifth negative bus bar **96***an* electrically connects the second junction **94** and one of the contact points **116** of the fifth negative contactor **78***e* to each other. The fifth negative bus bar **96***a* extends from the second junction **94** in the arrow Y2 direction.

The fifth negative bus bar **96***b* electrically connects the other of the contact points **116** of the fifth negative contactor **78***e* and the fifth negative connection terminal **46***e* to each other. The fifth negative bus bar **96***b* extends from the fifth negative contactor **78***e* in the arrow Y2 and arrow X2 direction. The fifth negative contactor **78***e* is positioned in the arrow X1 direction with respect to the fourth negative contactor **78***d*. The fifth negative contactor **78***e* is positioned on the back side of the second heat dissipation plate portion **134** of the second junction **86**. As shown in FIG. **7**, the fifth negative connection terminal **46***e* is disposed on the second side surface **108***b* of the support board **100**. The fifth negative connection terminal **46***e* is adjacent to the fifth positive connection terminal **44***e* in the arrow Y1 direction.

As shown in FIG. **5**, the sixth negative conductor **98** includes two sixth negative bus bars **98***a* and **98***b* as the second bus bars **113**. The sixth negative bus bar **98***an* electrically connects the second junction **94** and one of the contact points **116** of the sixth negative contactor **78***f* to each other. The sixth negative bus bar **98***a* is positioned in the arrow X1 direction with respect to the fifth negative bus bar **96***a*. The sixth negative bus bar **98***a* extends from the second junction **94** in the arrow Y2 direction. The sixth negative contactor **78***f* is positioned in the arrow X1 direction with respect to the fifth negative contactor **78***e*.

The sixth negative bus bar **98***b* electrically connects the other of the contact points **116** of the sixth negative contactor **78***f* and the sixth negative connection terminal **46***f* to each other. The sixth negative bus bar **98***b* extends from the sixth negative contactor **78***f* in the arrow Y2 and arrow X2 direction. As shown in FIG. **7**, the sixth negative connection terminal **46***f* is disposed on the second side surface **108***b* of the support board **100**. The sixth negative connection terminal **46***f* is adjacent to the sixth positive connection terminal **44***f* in the arrow Y2 direction.

As shown in FIGS. **6** and **7**, in the electric power device **10**, the first set of terminals **52***a* and the second set of terminals **52***b* are arranged on the first side surface **108***a* of the support board **100**, and the third set of terminals **52***c*, the fourth set of terminals **52***d*, the fifth set of terminals **52***e* and the sixth set of terminals **52***f* are arranged on the second side surface **108***b* of the support board **100**. Hereinafter, the first to sixth sets of terminals **52***a* to **52***f* may be simply referred to as sets of terminals **52**.

As shown in FIGS. **4** to **7**, an insulation projection **156** is provided between adjacent sets of terminals **52** for electrically insulating the sets of terminals **52** from each other. Further, for each set of terminals **52**, an insulation projection **158** is provided between the positive connection terminal **44** and the negative connection terminal **46** for electrically separating the positive connection terminal **44** and the negative connection terminal **46**. As shown in FIG. **6**, the distance L2 between the adjacent sets of terminals **52** is larger than the distance L1 between the positive connection terminal **44** and the negative connection terminal **46**.

According to the present embodiment, the first circuit portion **110** is disposed on the first surface **106***a* of the support board **100**, and the second circuit portion **112** is disposed on the second surface **106***b* of the support board **100**, whereby the first circuit portion **110** and the second circuit portion **112** are electrically separated from each other by the support board **100**. The support board **100** has electrical insulation properties and a certain thickness. Therefore, the positive electrode and the negative electrode of the direct current circuit **38** can be reliably insulated from each other with a simple configuration. The first and second sets of terminals **52***a*, **52***b* are disposed on the first side surface **108**_a_, and the third to sixth sets of terminals **52**_c_ to **52**_f_ are disposed on the second side surface **108**_b_. In this way, it is possible to gather cables to be connected to the electric power device **10** together. Therefore, the electric power device **10** can be prevented from becoming large.

In the electric power device **10**, as shown in FIG. **10**, the through hole **128** of the support board **100** may be blocked by a wall portion **160**. In this case, the ammeter **64** can be attached in a vertically inverted state.

In relation to the above-described disclosure, the following supplementary notes are further disclosed.

APPENDIX 1

The electric power device (**10**) including the direct current circuit (**38**) and the support board (**100**) having electrical insulation properties and configured to support the direct current circuit, wherein the direct current circuit including the positive connection terminal (**44**) and the negative connection terminal (**46**), each configured to be connected to an external circuit; the first circuit portion (**110**) including the first bus bar (**111**) electrically connected to one of the positive connection terminal or the negative connection terminal; and the second circuit portion (**112**) including a second bus bar (**113**) electrically connected to another of the positive connection terminal or the negative connection terminal, the first circuit portion disposed on the first surface (**106**_a_) of the support board and the second circuit portion disposed on the second surface (**106**_b_) of the support board are electrically separated from each other by the support board, and the set (**52**) of the positive connection terminal and the negative connection terminal which are paired is arranged on the same side surface (**108**_a_, **108**_b_) of the support board.

In accordance with such a configuration, the first circuit portion is disposed on the first surface of the support board, and the second circuit portion is disposed on the second surface of the support board, and thus it is possible to reliably insulate the positive electrode and the negative electrode of the direct current circuit with a simple configuration. Further, since the set of terminals is disposed on the same side surface of the support board, it is possible to gather cables to be connected to the electric power device. This can suppress an increase in the size of the electric power device.

APPENDIX 2

In the electric power device according to Appendix 1, the set of the terminals may include multiple sets of the terminals, and the multiple sets of the terminals may be arranged on different side surfaces of the support board.

According to such a configuration, since multiple sets of terminals are arranged on different side surfaces, the effect of insulation of the sets of terminals can be enhanced.

APPENDIX 3

In the electric power device according to Appendix 2, the different side surfaces on which the multiple sets of the terminals are arranged may be opposite side surfaces of the support board facing in opposite directions.

According to such a configuration, the effect of insulation of the plurality of sets of terminals can be further enhanced. In addition, it is possible to suppress cables connected to the plurality of sets of terminals from being entangled with each other.

APPENDIX 4

In the electric power device according to any one of the Appendices 1 to 3, the set of the terminals may be the multiple sets of the terminals, the multiple sets of the terminals are arranged on the same side surface, and the distance (L2) between the multiple sets of the terminals arranged adjacent to each other may be larger than the distance (L1) between the positive connection terminal and the negative connection terminal forming each of the adjacent two of the polarity of sets of terminals.

In accordance with such a configuration, it is possible to suppress erroneous assembly of cables to be connected to the sets of terminals while insulating the sets of terminal arranged adjacent to each other.

APPENDIX 5

In the electric power device according to any one of Appendices 1 to 4, in the first circuit portion, the first bus bar may be provided with a plurality of contactors (**62**), and the plurality of contactors may be arranged along the side surface of the support board on which the set of terminals is arranged.

In accordance with such a configuration, the length of the first bus bar between the connection terminal and the contactor can be shortened.

APPENDIX 6

In the electric power device according to any one of the Appendices 1 to 5, the first circuit portion may include at least three of the first bus bars and a plurality of junctions (**56**, **86**) to which the at least three first bus bars are electrically connected, the plurality of junctions each having a plate shape, and the plurality of junctions may be disposed on the support board in a state of being separated from each other.

In accordance with such a configuration, heat can be efficiently dissipated by the plate-shaped junctions. Further, since the plurality of junctions are separated from each other, the plurality of junctions can be insulated.

APPENDIX 7

In the electric power device according to any one of Appendices 1 to 6, the first circuit portion may include at least three of the first bus bars and a junction to which the at least three first bus bars are connected, the junction having a plate shape, the second circuit portion may include a contactor (**78**) provided on the second bus bar, the support board may include a first fixing portion (**150**) to which the junction is fixed and a second fixing portion (**152**) to which the contactor is fixed, and the first fixing portion may be adjacent to the second fixing portion.

In accordance with such a configuration, the second fixing portion supporting the contactor can be reinforced by the plate-shaped junction. This can suppress the support board from being bent by the contactor.

APPENDIX 8

A moving object (**12**) includes the electric power device according to any one of Appendixes 1 to 7.

Moreover, it should be noted that the present invention is not limited to the disclosure described above, and various configurations may be adopted therein without departing from the essence and gist of the present invention.

The invention claimed is:

1. An electric power device, comprising:

a direct current circuit; and

a support board having electrical insulation properties and configured to support the direct current circuit, wherein

the direct current circuit comprises:

a positive connection terminal and a negative connection terminal, each configured to be connected to an external circuit;

a first circuit portion including a first bus bar electrically connected to one of the positive connection terminal or the negative connection terminal; and

a second circuit portion including a second bus bar electrically connected to another of the positive connection terminal or the negative connection terminal,

the first circuit portion disposed on a first surface of the support board and the second circuit portion disposed on a second surface of the support board are electrically separated from each other by the support board, and

a set of the positive connection terminal and the negative connection terminal which are paired is arranged on a same side surface of the support board.

2. The electric power device according to claim **1**, wherein

the set of the terminals includes multiple sets of the terminals, and

the multiple sets of the terminals are arranged on different side surfaces of the support board.

3. The electric power device according to claim **2**, wherein

the different side surfaces on which the multiple sets of terminals are arranged are opposite side surfaces of the support board facing in opposite directions.

4. The electric power device according to claim **1**, wherein

the set of the terminals includes multiple sets of the terminals,

the multiple sets of terminals are arranged on the same side surface, and

a distance between the multiple sets of the terminals arranged adjacent to each other is larger than a distance between the positive connection terminal and the negative connection terminal forming each of the multiple sets of the terminals.

5. The electric power device according to claim **1**, wherein

in the first circuit portion, the first bus bar is provided with a plurality of contactors, and

the plurality of contactors are arranged along the side surface of the support board on which the set of terminals is arranged.

6. The electric power device according to claim **1**, wherein

the first circuit portion includes at least three of the first bus bars and a plurality of junctions to which the at least three first bus bars are electrically connected, the plurality of junctions each having a plate shape, and

the plurality of junctions are disposed on the support board in a state of being separated from each other.

7. The electric power device according to claim **1**, wherein

the first circuit portion includes at least three of the first bus bars and a junction to which the at least three first bus bars are connected, the junction having a plate shape,

the second circuit portion includes a contactor provided on the second bus bar,

the support board includes:

a first fixing portion to which the junction is fixed; and

a second fixing portion to which the contactor is fixed, and

the first fixing portion is adjacent to the second fixing portion.

8. A moving object comprising the electric power device according to claim **1**.

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